FUNCTORIALITY OF CLOSED SUBSETS PROPER OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: all three clauses are proved by reducing to reduced induced closed subspaces and quoting the standard Morphisms of Spaces lemmas for base change, composition and images; every clause is written out in full

Editorial note. The label lemma-functoriality-closed-proper-over-base names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- topology on the set of points of an algebraic space and reduced induced closed subspace structures
- proper, universally closed and separated morphisms of algebraic spaces and their behaviour under base change, composition and closed immersions
- the standard morphism lemmas of Spaces quoted by the source (scheme-theoretic image of a proper morphism, image of universally closed over separated target, image of proper is proper, immersion locally of finite type, composition of finite type)

Source reference. Original lemma 04WP: sources/stacks-chow/tags.txt, source lines 6367–6367. Prerequisite pointer invoked at proof line(s) 1175; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04XZ: sources/stacks-chow/tags.txt, source lines 6412–6412. Prerequisite pointer invoked at proof line(s) 1179; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03XG: sources/stacks-chow/tags.txt, source lines 5147–5147. Prerequisite pointer invoked at proof line(s) 1190; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04XY: sources/stacks-chow/tags.txt, source lines 6411–6411. Prerequisite pointer invoked at proof line(s) 1180; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08AJ: sources/stacks-chow/tags.txt, source lines 10534–10534. Prerequisite pointer invoked at proof line(s) 1194; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05Z2: sources/stacks-chow/tags.txt, source lines 7676–7676. Prerequisite pointer invoked at proof line(s) 1183; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06ED: sources/stacks-chow/tags.txt, source lines 8212–8212. Prerequisite pointer invoked at proof line(s) 1189; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AGD: sources/stacks-chow/tags.txt, source lines 13191–13191. Prerequisite pointer invoked at proof line(s) 1164; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $f : X \to Y$ be a morphism of algebraic spaces which are locally of finite type over B.*

1. *If Y is separated over B and $T \subset |X|$ is a closed subset proper over B, then $|f|(T)$ is a closed subset of $|Y|$ proper over B.*
2. *If f is universally closed and $T \subset |X|$ is a closed subset proper over B, then $|f|(T)$ is a closed subset of Y proper over B.*
3. *If f is proper and $T \subset |Y|$ is a closed subset proper over B, then $|f|^{-1}(T)$ is a closed subset of $|X|$ proper over B.*

Proof. Proof of (1). Assume Y is separated over B and $T \subset |X|$ is a closed subset proper over B. Let Z be the reduced induced closed subspace structure on T and apply Morphisms of Spaces, Lemma 0AGD to $Z \to Y$ over B to conclude.

Proof of (2). Assume f is universally closed and $T \subset |X|$ is a closed subset proper over B. Let Z be the reduced induced closed subspace structure on T and let Z' be the reduced induced closed subspace structure on $|f|(T)$. We obtain an induced morphism $Z \to Z'$. Denote $Z'' = f^{-1}(Z')$ the scheme theoretic inverse image. Then $Z'' \to Z'$ is universally closed as a base change of f (Morphisms of Spaces, Lemma 04WP). Hence $Z \to Z'$ is universally closed as a composition of the closed immersion $Z \to Z''$ and $Z'' \to Z'$ (Morphisms of Spaces, Lemmas 04XZ and 04XY). We conclude that $Z' \to B$ is separated by Morphisms of Spaces, Lemma 05Z2. Since $Z \to B$ is quasi-compact and $Z \to Z'$ is surjective we see that $Z' \to B$ is quasi-compact. Since $Z' \to B$ is the composition of $Z' \to Y$ and $Y \to B$ we see that $Z' \to B$ is locally of finite type (Morphisms of Spaces, Lemmas 06ED and 03XG). Finally, since $Z \to B$ is universally closed, we see that the same thing is true for $Z' \to B$ by Morphisms of Spaces, Lemma 08AJ. This finishes the proof.

Proof of (3). Assume f is proper and $T \subset |Y|$ is a closed subset proper over B. Let Z be the reduced induced closed subspace structure on T. Denote $Z' = f^{-1}(Z)$ the scheme theoretic inverse image. Then $Z' \to Z$ is proper as a base change of f (Morphisms of Spaces, Lemma 04WP). Whence $Z' \to B$ is proper as the composition of $Z' \to Z$ and $Z \to B$ (Morphisms of Spaces, Lemma 04XY). This finishes the proof. □